

The Universe as a Discovery Factory

Center for Computational Astrophysics (CCA)

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4/16/2025

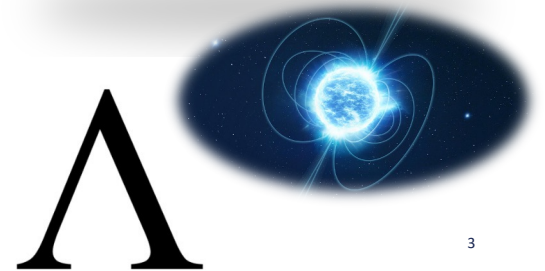
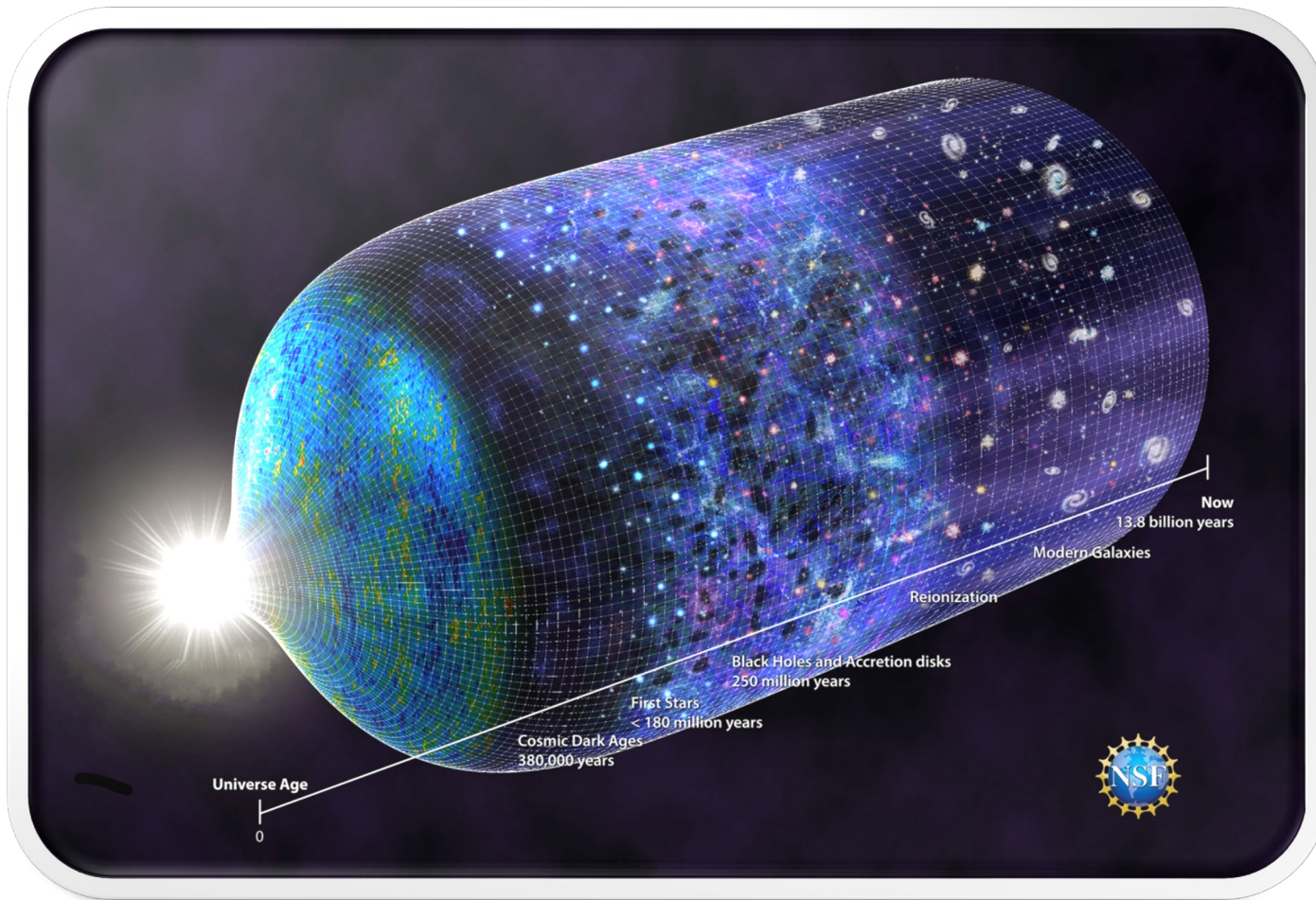


The Edge of Physics

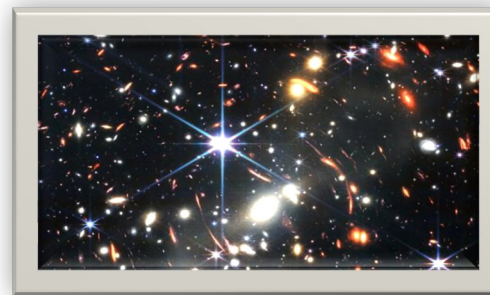
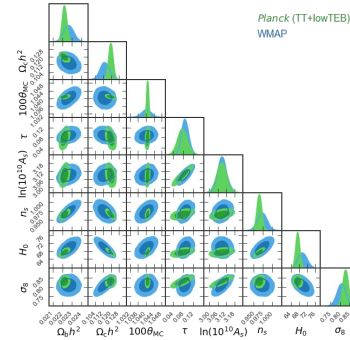
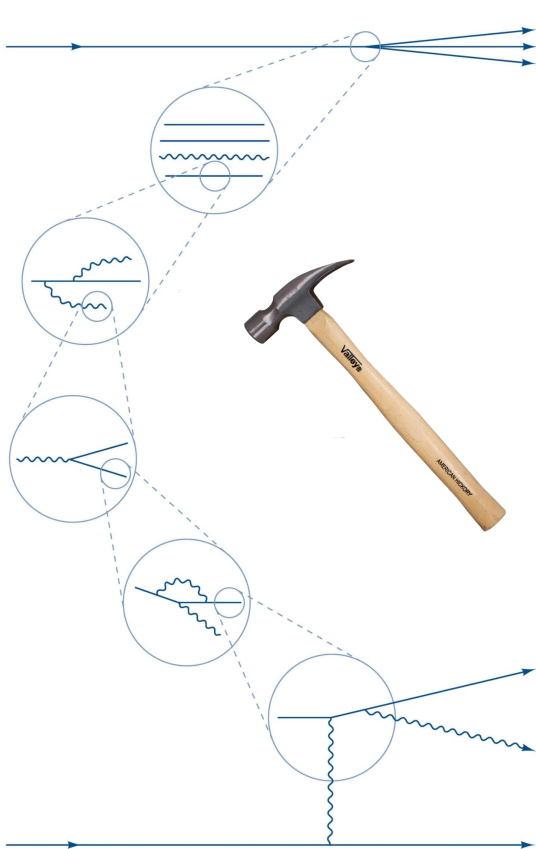


[CAMELS, Kaze Wong (2023)]

The Universe and its Constituents



The Universe for Fundamental Physics

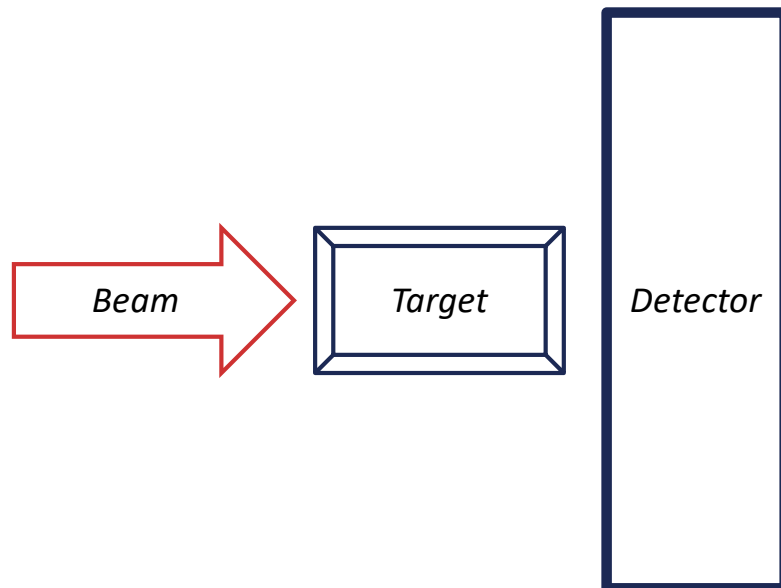


New
Physics

I. Discovery

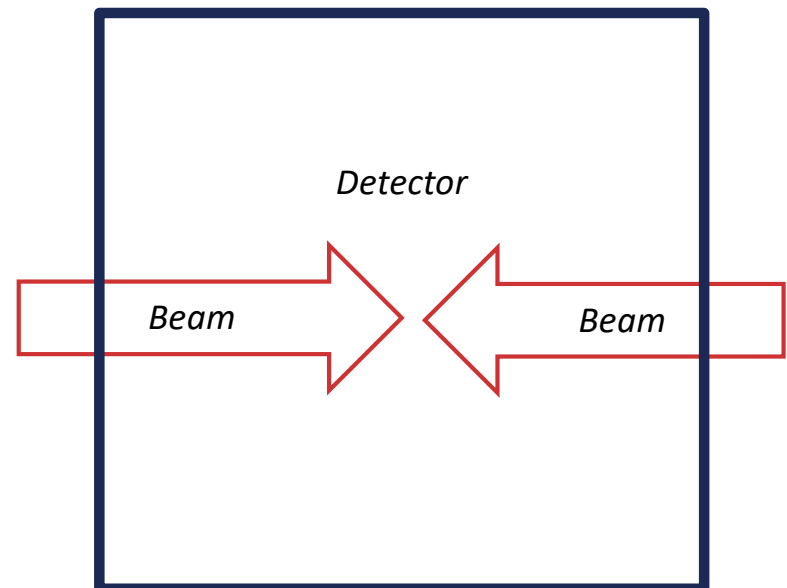
Discovery in Fundamental Physics

Fixed-Target



e.g. Rutherford's Gold Foil Experiment
and the discovery of the atomic nucleus

Collider



e.g. Large Hadron Collider (LHC)
and the discovery of the Higgs Boson

Current Status

Standard Model of Elementary Particles

three generations of matter (fermions)			interactions / force carriers (bosons)	
			I	II
mass	$\approx 2.16 \text{ MeV}/c^2$	$\approx 1.273 \text{ GeV}/c^2$	$\approx 172.57 \text{ GeV}/c^2$	$\approx 125.2 \text{ GeV}/c^2$
charge	$\frac{2}{3}$	$\frac{2}{3}$	$\frac{2}{3}$	0
spin	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	0
QUARKS	u up	c charm	t top	g gluon
	$\approx 4.7 \text{ MeV}/c^2$	$\approx 93.5 \text{ MeV}/c^2$	$\approx 4.183 \text{ GeV}/c^2$	$\approx 125.2 \text{ GeV}/c^2$
	$-\frac{1}{3}$	$-\frac{1}{3}$	$-\frac{1}{3}$	0
	d down	s strange	b bottom	γ photon
	$\approx 0.511 \text{ MeV}/c^2$	$\approx 105.66 \text{ MeV}/c^2$	$\approx 1.77693 \text{ GeV}/c^2$	$\approx 91.188 \text{ GeV}/c^2$
	-1	-1	-1	0
	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	1
LEPTONS	e electron	μ muon	τ tau	Z Z boson
	$< 0.8 \text{ eV}/c^2$	$< 0.17 \text{ MeV}/c^2$	$< 18.2 \text{ MeV}/c^2$	$\approx 80.3692 \text{ GeV}/c^2$
	0	0	0	0
	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	± 1
	ν_e electron neutrino	ν_μ muon neutrino	ν_τ tau neutrino	W W boson
	$< 0.8 \text{ eV}/c^2$	$< 0.17 \text{ MeV}/c^2$	$< 18.2 \text{ MeV}/c^2$	$\approx 80.3692 \text{ GeV}/c^2$
	0	0	0	0
	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	± 1

SCALAR BOSONS

GAUGE BOSONS
VECTOR BOSONS

Outstanding Questions

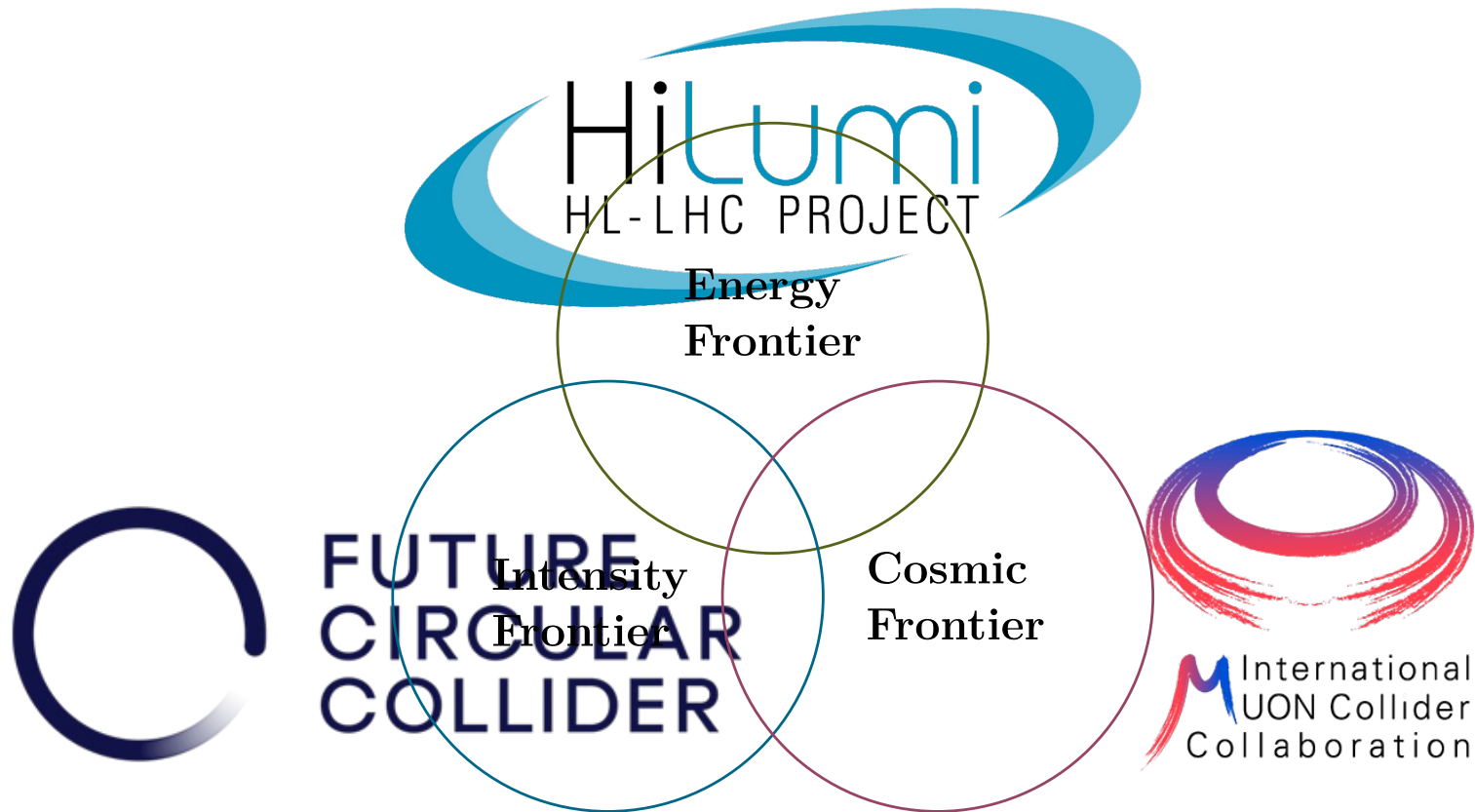
- Gravity?
- Why Three Generations?
- Why is the Higgs so light?
- Neutrino Masses?
- Why Matter over Anti-Matter?
- Dark Matter? Dark Energy?
- Origin of the Universe?

...

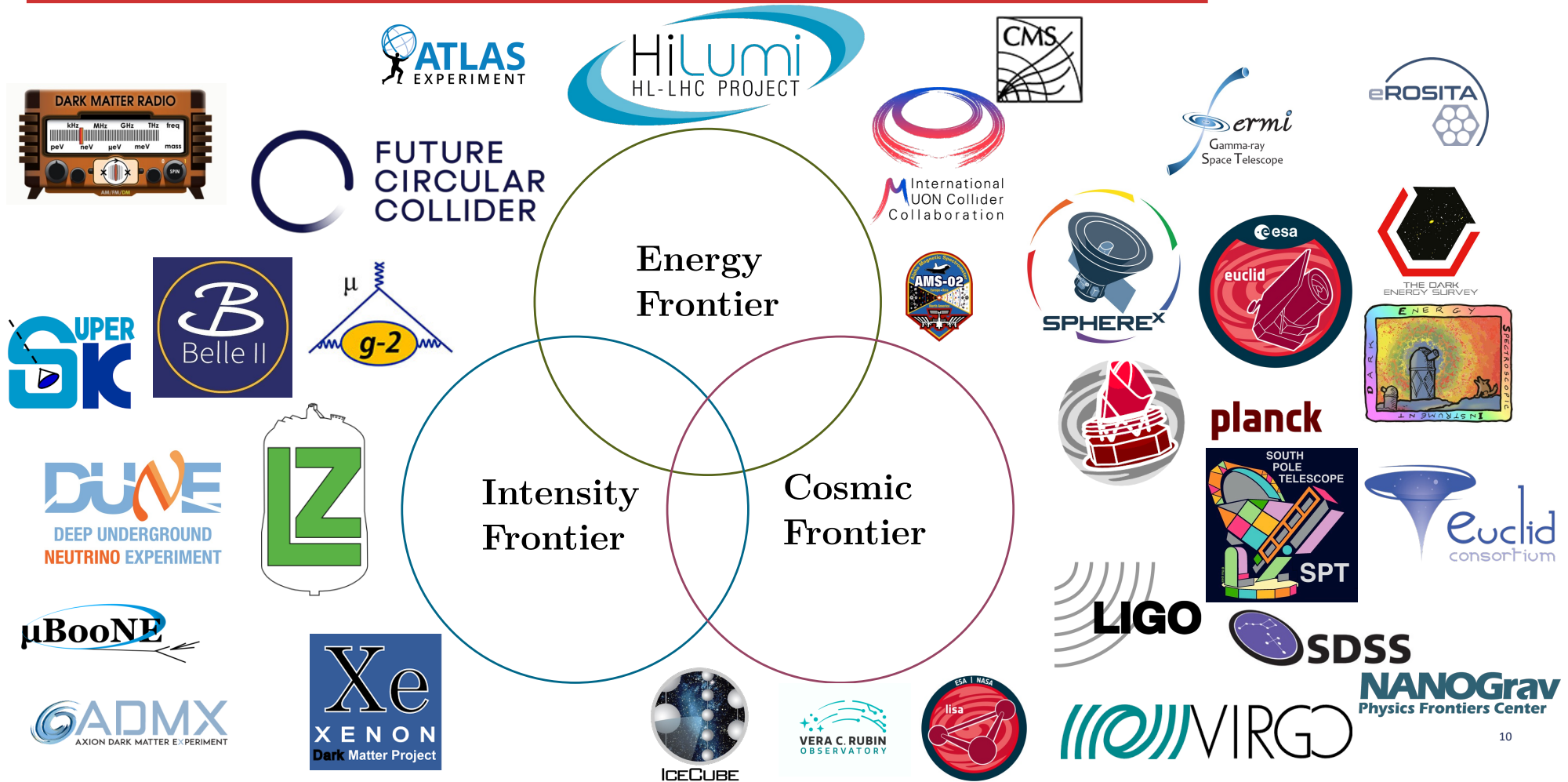
Future Discovery in Fundamental Physics



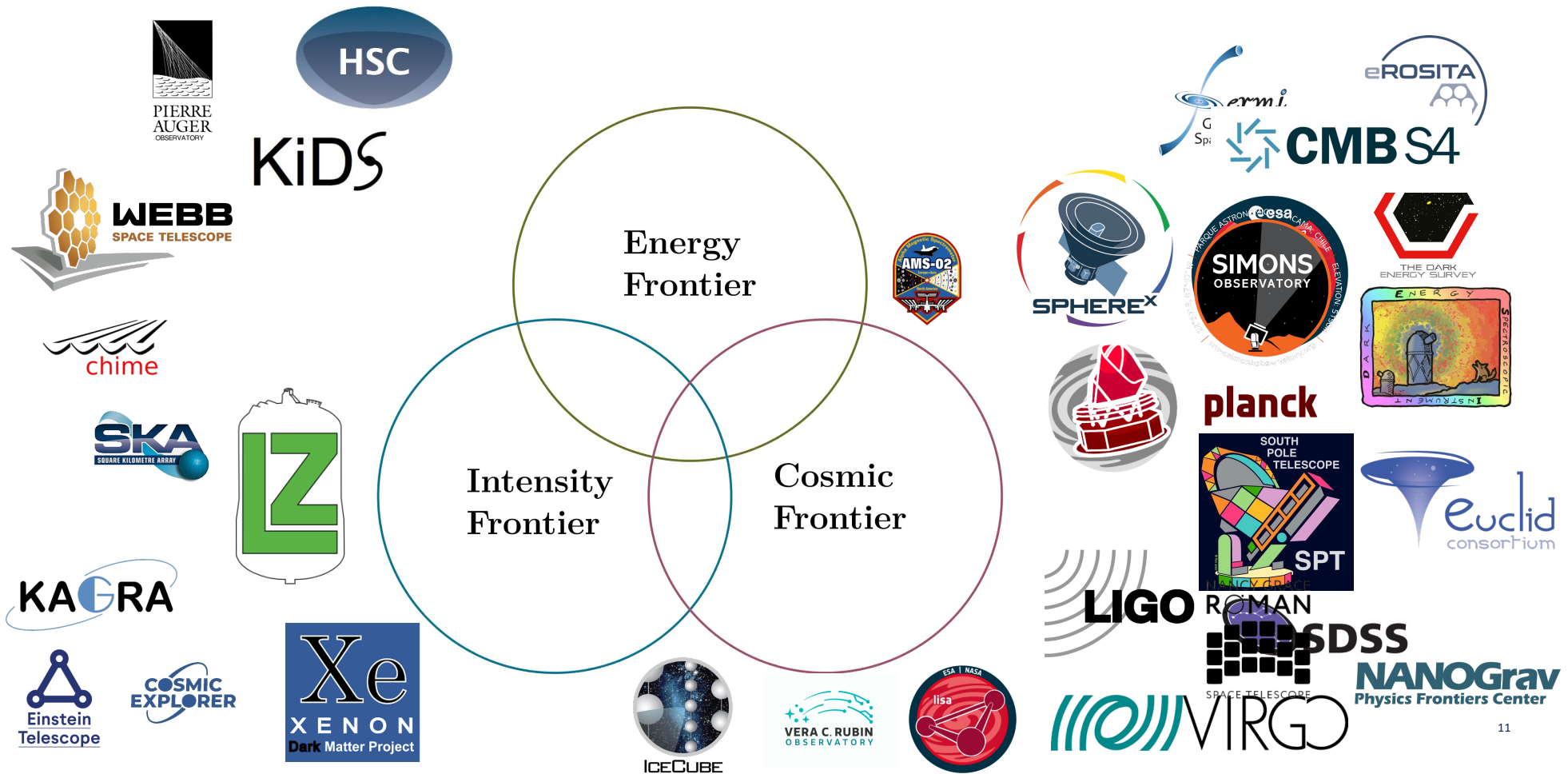
Future Discovery in Fundamental Physics



Future Discovery in Fundamental Physics



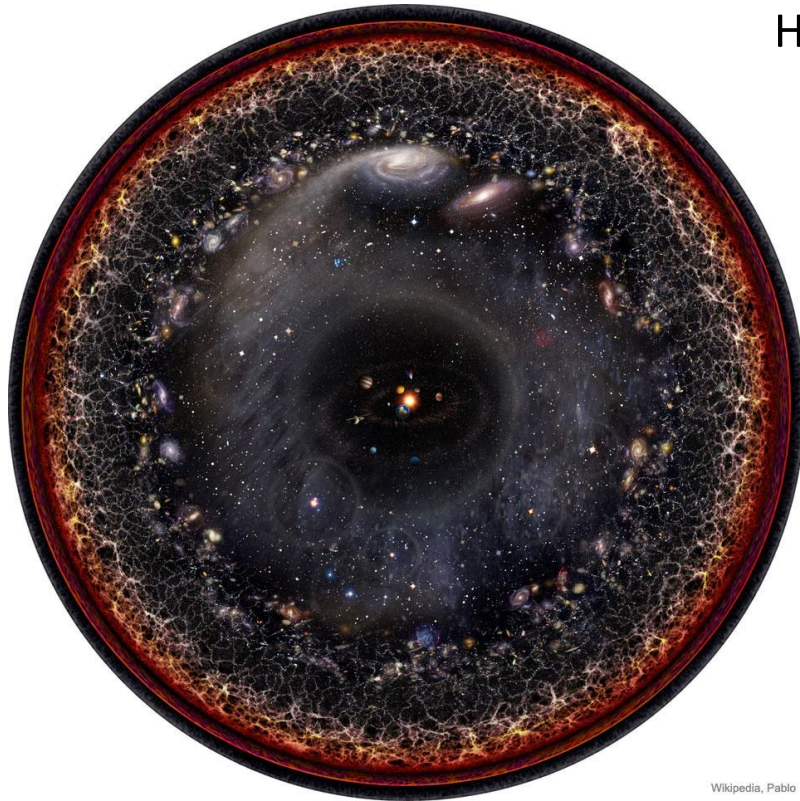
Future Discovery in Fundamental Physics



II. The Universe

The (Observable) Universe

Homogeneous and Isotropic



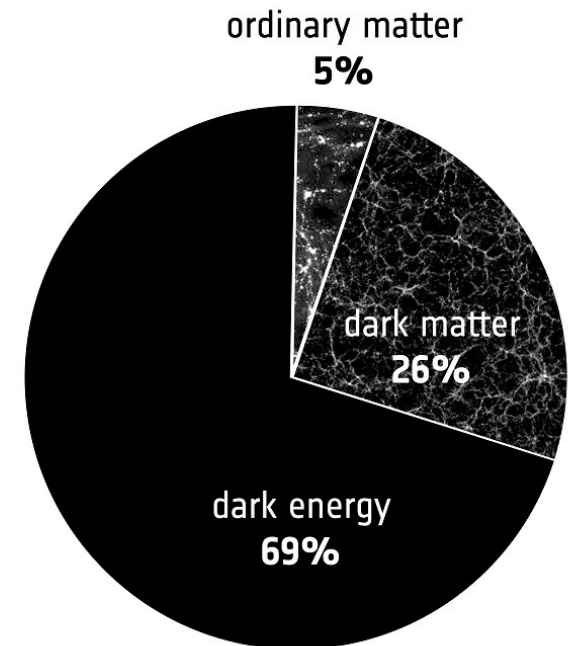
Wikipedia, Pablo

96 Billion Lightyears
 $\sim 10^{27}$ m

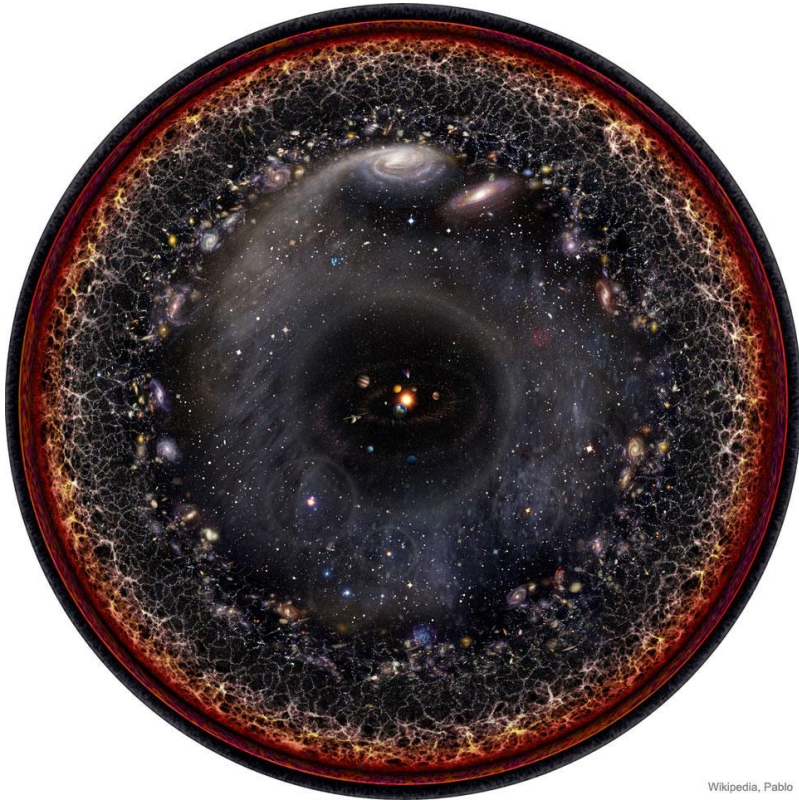
Diameter of Milky Way:
0.1 Million Lightyears $\sim 10^{21}$ m
(Million Times Smaller)

Age: 13.8 Billion Years

Earth's Age: 4.5 Billion Years ($\sim 1/3$ Universe's)



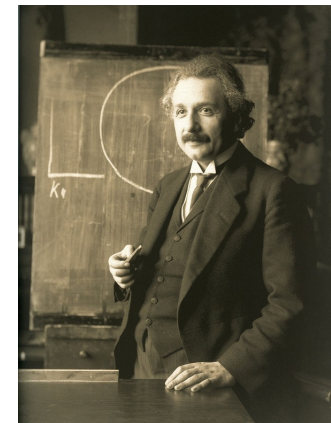
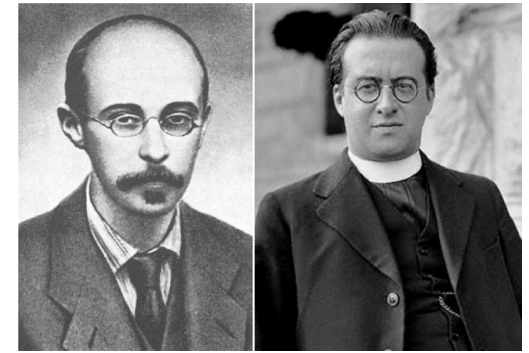
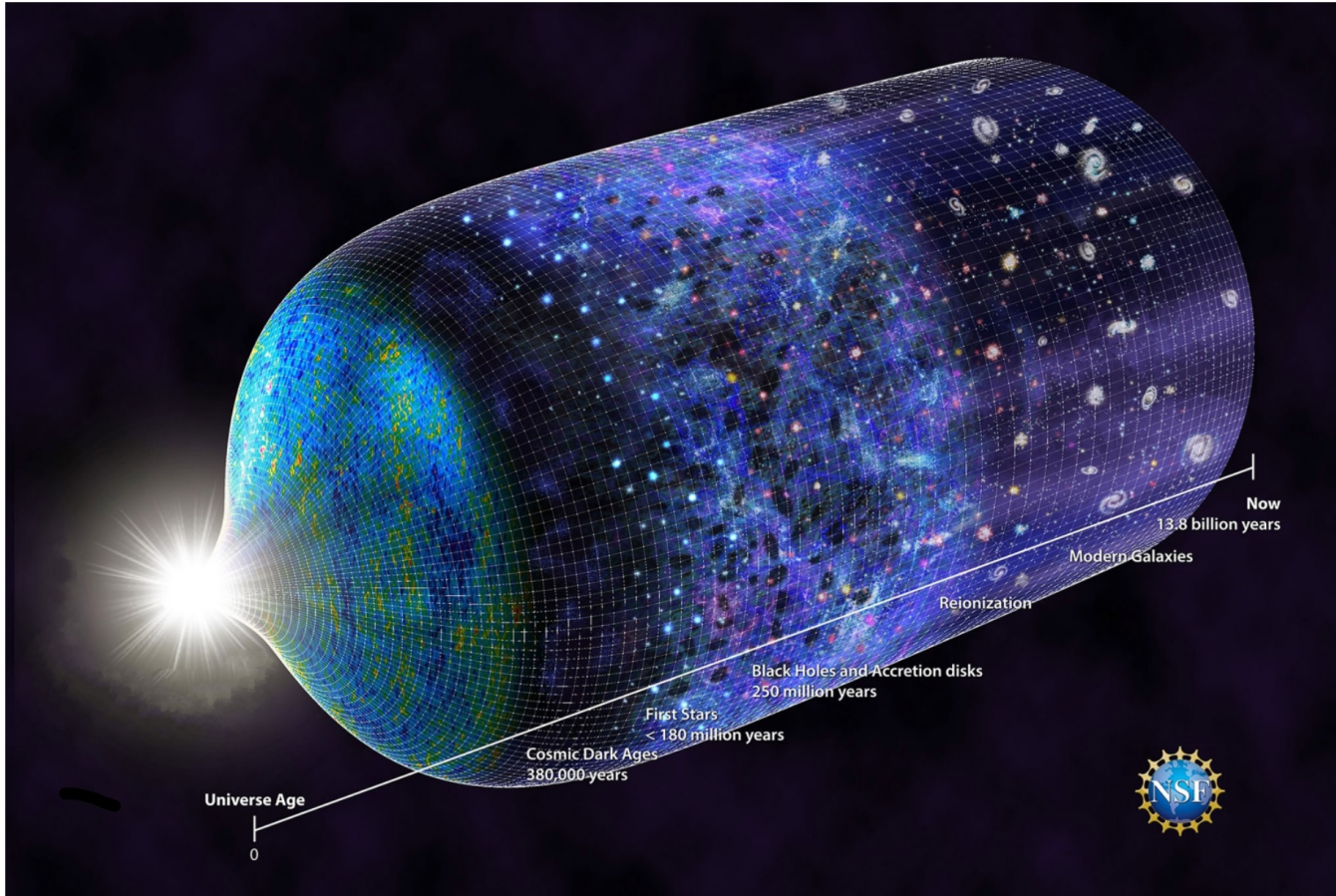
The Universe as a Laboratory?



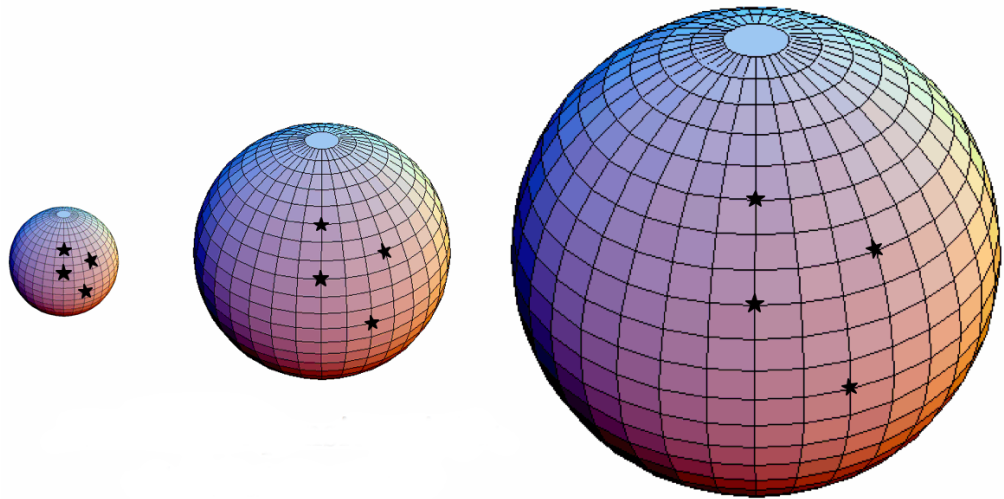
Wikipedia, Pablo

- 1) The Universe was once small
- 2) New Physics can hide at large scales

A Dynamical Universe

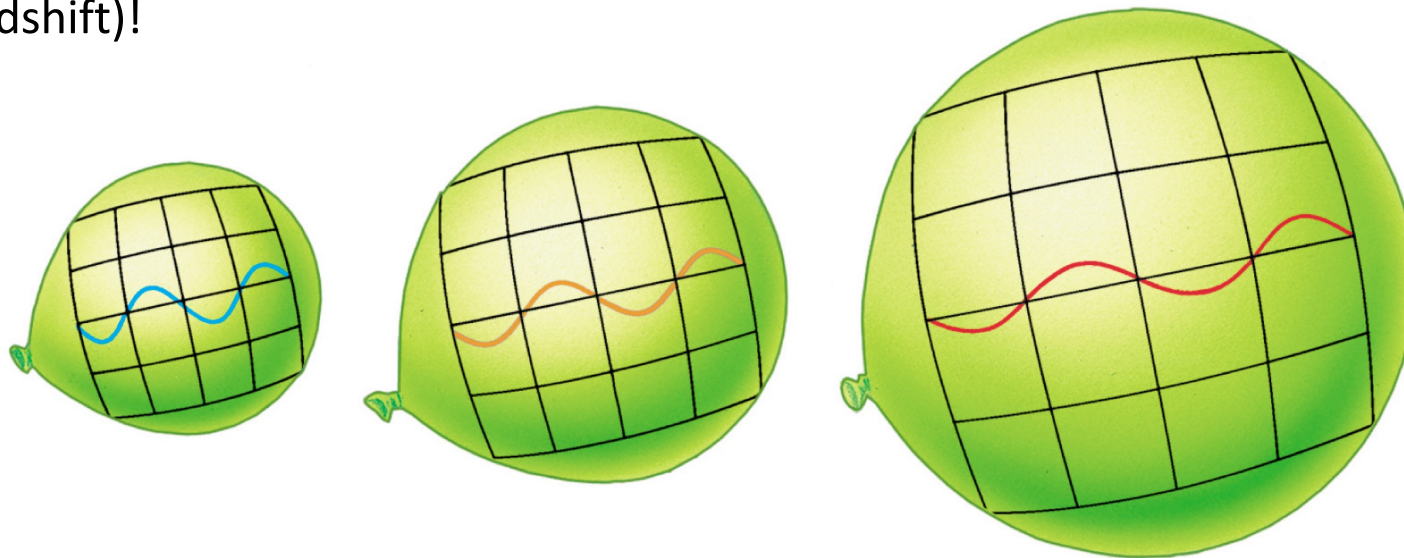


What does it mean for a Universe to expand?



What does it mean for a Universe to expand?

While point particles remain unchanged, the wavelength of waves expand (redshift)!



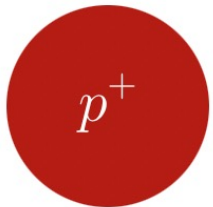
Two Predictions of an Expanding Universe

- 1) A Singularity (often called 'The Big Bang') [Hence, we can calculate age/size.]
 - a) Inflation? Ekpyrotic Universe?
Quantum Gravity? Chaotic BKL Conditions?
Other Jargon?
Observationally Unknown
- 2) Hot Big Bang (When the Universe was 1 second old)
 - a) Big Bang Nucleosynthesis (BBN) : Light Element Production
 - b) Cosmic Microwave Background (CMB) : Thermal Afterglow

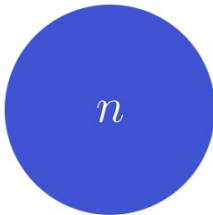
III. Observables

Big Bang Nucleosynthesis (BBN)

$t \gtrsim 1 \text{ sec}$



Protons (p)



Neutrons (n)



Electrons (e)



Neutrinos (ν)



Photons (γ)

At 1 second Weak Interactions

$$\nu_e + \bar{\nu}_e \leftrightarrow e^+ + e^-$$

$$e^- + \bar{\nu}_e \leftrightarrow e^- + \bar{\nu}_e$$

$$n + \nu_e \leftrightarrow p^+ + e^-$$

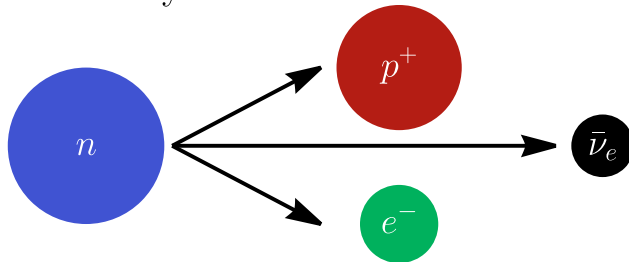
$$n + e^+ \leftrightarrow p^+ + \bar{\nu}_e$$

Turn off

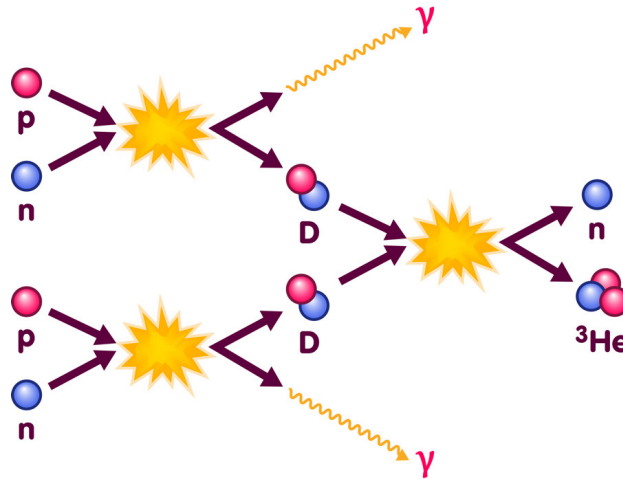
Big Bang Nucleosynthesis (BBN)

$$t \gtrsim 100 \text{ sec}$$

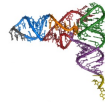
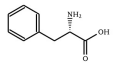
Beta Decay



$$t_{\text{nuc}} \simeq 250 \text{ sec}$$



$$\frac{4n_{\text{He}}}{n_{\text{H}}} \sim \frac{1}{4}$$



Group 1

1

2.00

1s

H

hydrogen

1.008

3

0.98

2s

Li

lithium

6.941

4

1.57

2p

Be

beryllium

9.012

Sy

element

size

2s

atomic number

1s

Filling electro-negativity

sc

last occupied subshell

Sc

symbol

element

element name

standard atomic weight

11

0.93

3s

Na

sodium

22.99

12

1.31

3s

Mg

magnesium

24.305

19

0.82

4s

K

potassium

40.078

20

1.00

4s

Ca

calcium

40.078

21

1.36

3d

Sc

scandium

44.956

22

1.54

3d

Ti

titanium

50.942

23

1.63

3d

V

vanadium

50.942

24

1.66

3s

Cr

chromium

51.996

25

1.55

3s

Mn

manganese

54.938

26

1.81

3d

Fe

iron

55.845

27

2.00

3d

Co

cobalt

58.933

28

1.91

3d

Ni

nickel

58.693

29

1.90

3d

Cu

copper

63.546

30

1.65

3d

Zn

zinc

65.38

31

1.81

4p

Ga

gallium

69.723

32

2.01

4p

Ge

germanium

72.64

33

2.18

4p

As

arsenic

74.922

34

2.55

4p

Se

selenium

78.971

35

3.46

4p

Br

bromine

79.904

36

4.00

4p

Kr

krypton

83.798

37

0.82

5s

Rb

rubidium

85.468

38

0.95

5s

Sr

strontium

87.62

39

1.22

4d

Y

yttrium

88.906

40

1.31

4d

Zr

zirconium

91.224

41

1.56

4d

Nb

niobium

92.906

42

1.64

4d

Mo

molybdenum

95.94

43

1.9

4d

Tc

technetium

(97)

44

2.2

4d

Ru

ruthenium

101.07

45

2.28

4d

Rh

rhodium

102.91

46

2.20

4d

Pd

palladium

106.42

47

1.93

4d

Ag

silver

107.87

48

1.69

4d

Cd

cadmium

112.41

49

1.78

5p

In

indium

114.82

50

1.96

5p

Sn

antimony

118.71

51

2.05

5p

Sb

tellurium

127.6

52

2.1

5p

I

iodine

126.91

53

2.66

5p

Xe

xenon

131.29

55

0.79

6s

Cs

caesium

132.91

56

0.89

6s

Ba

barium

137.33

57

1.38

5d

Lanthanides

72

1.3

5f

Hf

hafnium

178.49

73

1.5

5f

Ta

tantalum

180.95

74

2.36

5f

W

wolfram

183.84

75

1.8

5f

Re

rhenium

186.21

76

2.2

5f

Os

osmium

190.23

77

2.2

5f

Ir

iridium

192.22

78

2.28

5f

Pt

platinum

195.08

79

2.54

5f

Au

gold

196.97

80

1.9

5f

Hg

mercury

200.59

81

1.82

6p

Tl

thallium

204.38

82

2.0

6p

Pb

lead

207.2

83

2.52

6p

Bi

bismuth

208.98

84

2.6

6p

Po

polonium

(209)

85

2.7

6p

At

astatine

(210)

86

2.8

6p

Rn

radon

(222)

87

Fr

francium

(223)

88

Ra

radium

(226)

**

actinides

104

Rf

rutherfordium

(261)

105

Db

dundubium

(263)

106

Sg

seaborgium

(266)

107

Bh

bohrium

(269)

108

Hs

hassium

(271)

109

Ds

darmstadtium

(285)

110

Rg

roentgenium

(288)

111

Nh

nihonium

(291)

112

Cn

copernicium

(285)

113

Nh

flerovium

(289)

114

Fl

flerovium

(289)

115

Mc

moscovium

(293)

116

Lv

livermorium

(293)

117

Ts

tennessine

(294)

118

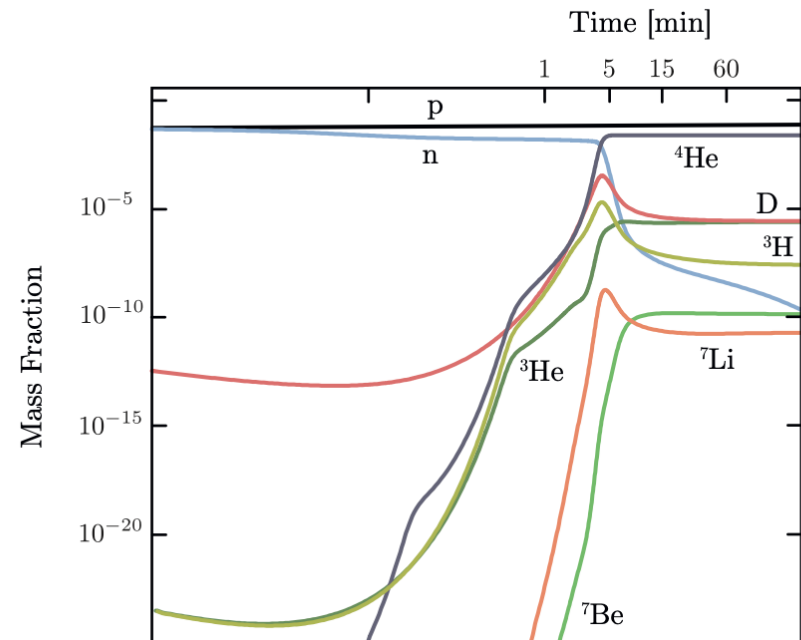
Og

oganeson

(294)

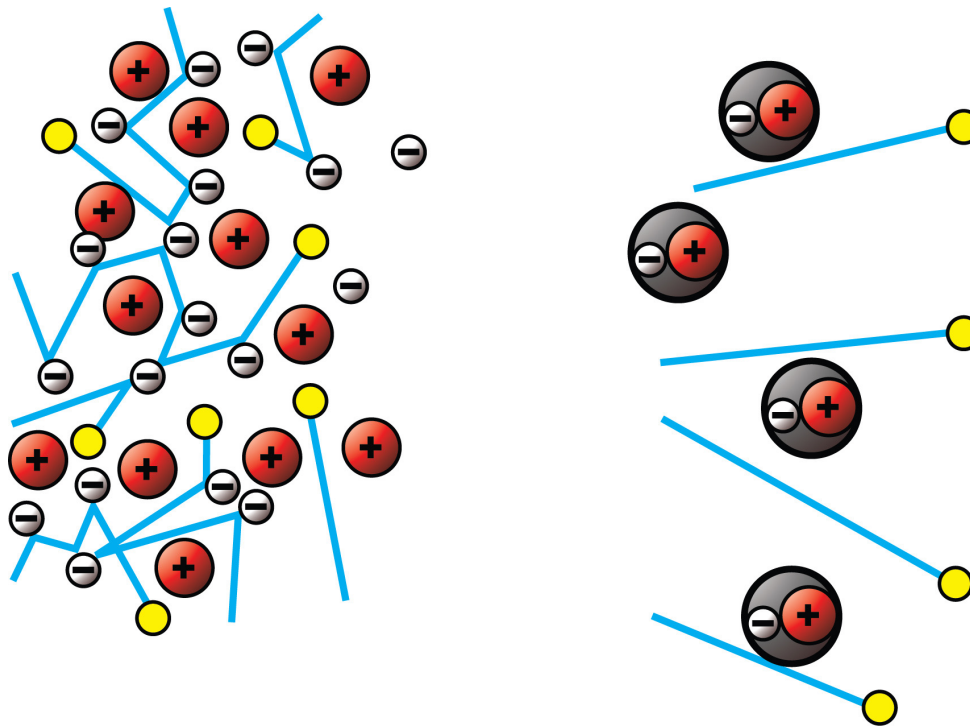
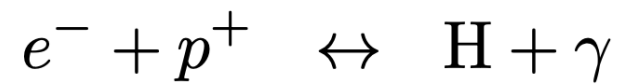
[illegible]

* Indicates an anomalous (Aufbau rule-breaking) ground state electron configuration.
 Inspired by Ivan Griffin's ElX Periodic Table. ElXcode is released under the MIT open source license.
 Final product (this Table) is released under creative commons attribution/share-alike copyright terms. © 2022. Paul N. Danese



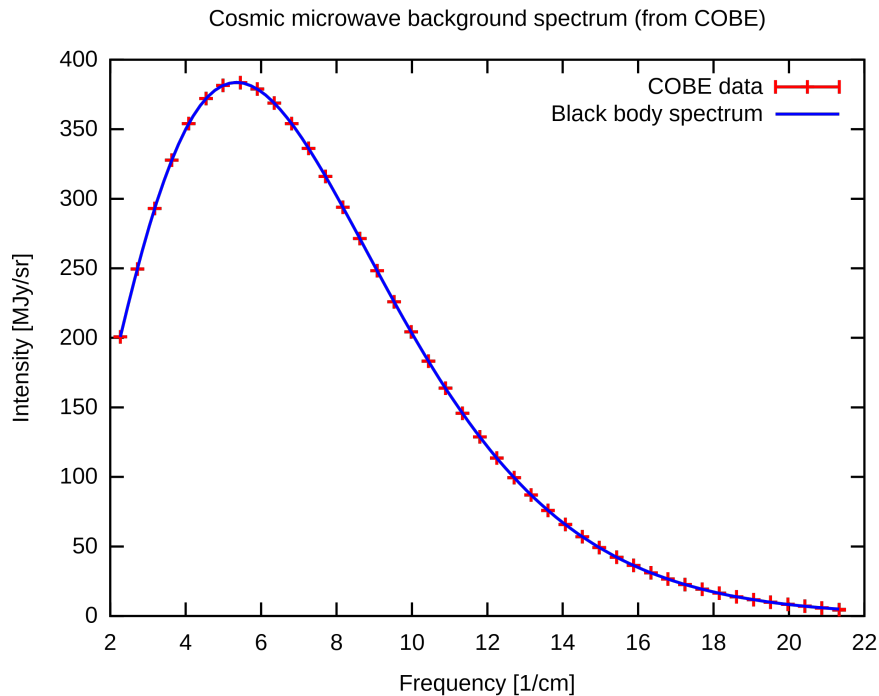
Recombination

380 kyr

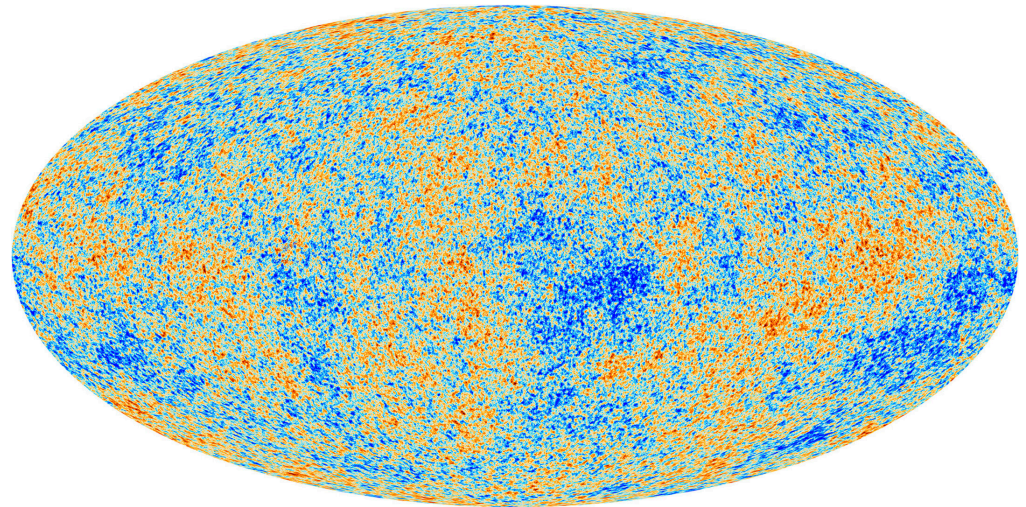


Free Streaming Photons = Cosmic Microwave Background (CMB)!

The Cosmic Microwave Background



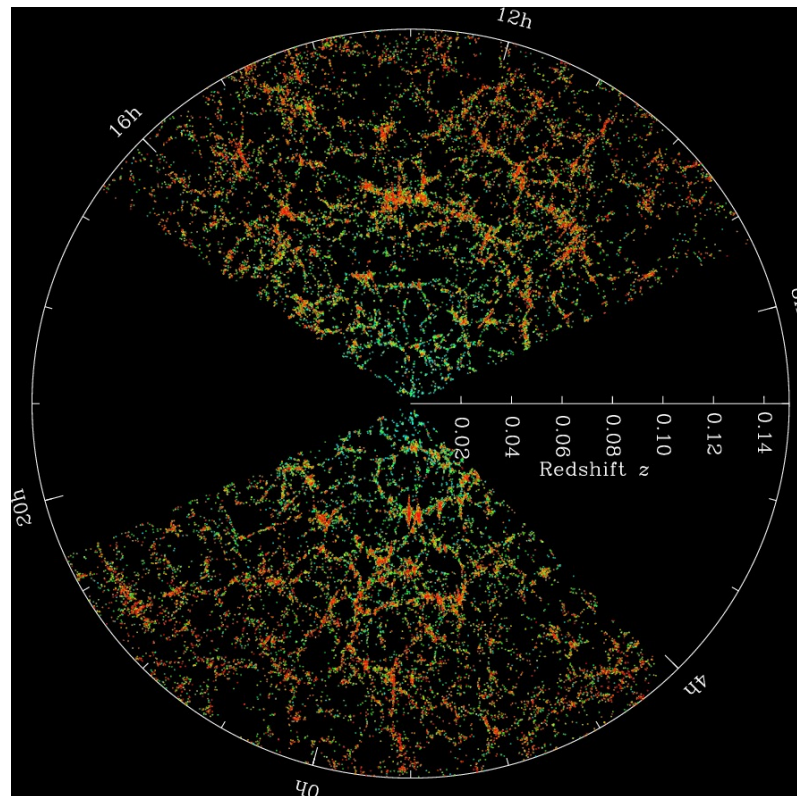
Very Isotropic!



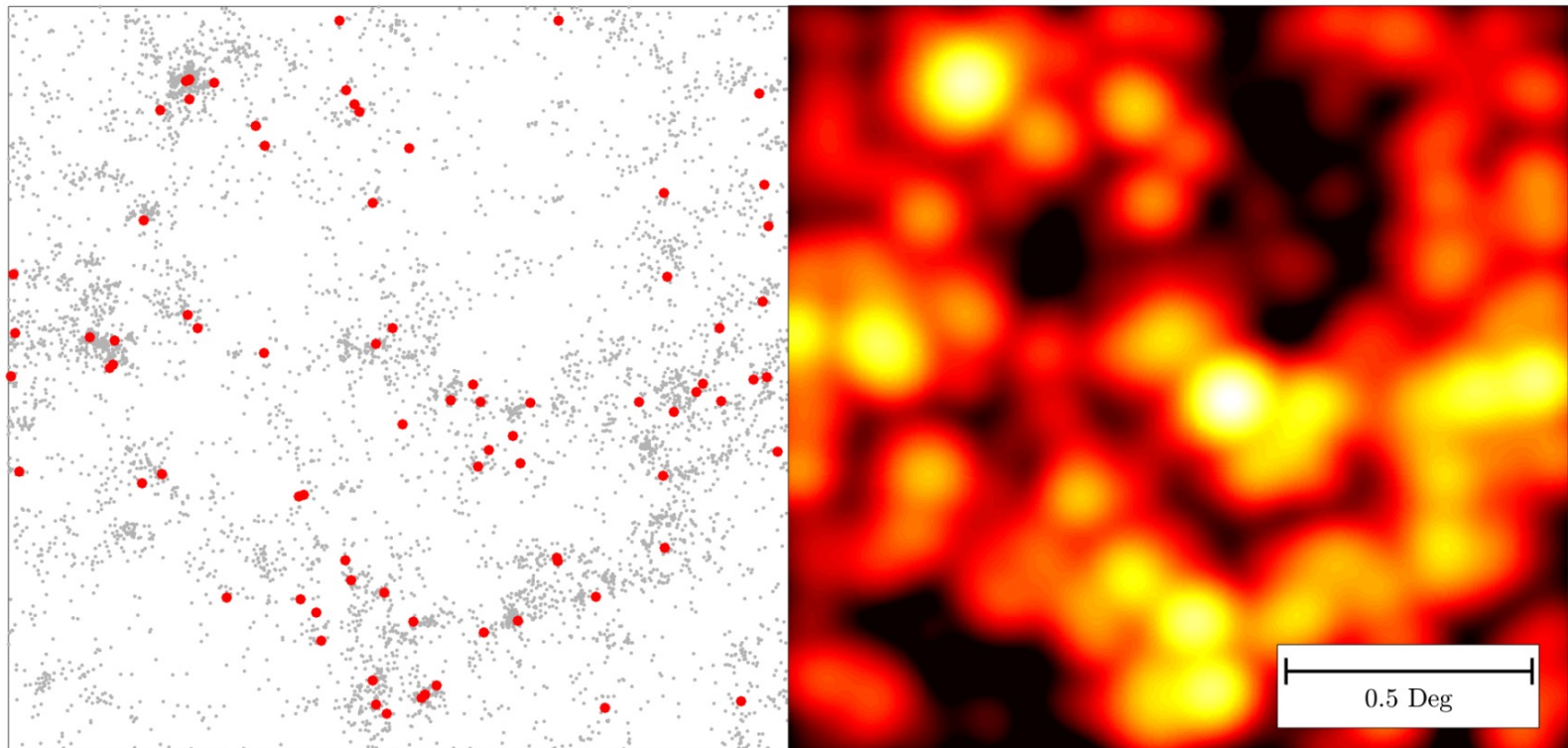
First discovered in 1965 by Penzias and Wilson

The Cosmic Background Explorer COBE Measured Spectrum 1993

Large-Scale Structure



Line-Intensity Mapping



Cosmological Observables Summary

- 1) BBN
- 2) CMB Spectrum
- 3) CMB Temperature
- 4) Large-Scale Structure
- 5) Line-Intensity Mapping

...

Cosmological Observables to Questions

1)BBN
2)CMB Spectrum
3)CMB Temperature
4)Large-Scale Structure
5) Line-Intensity Mapping

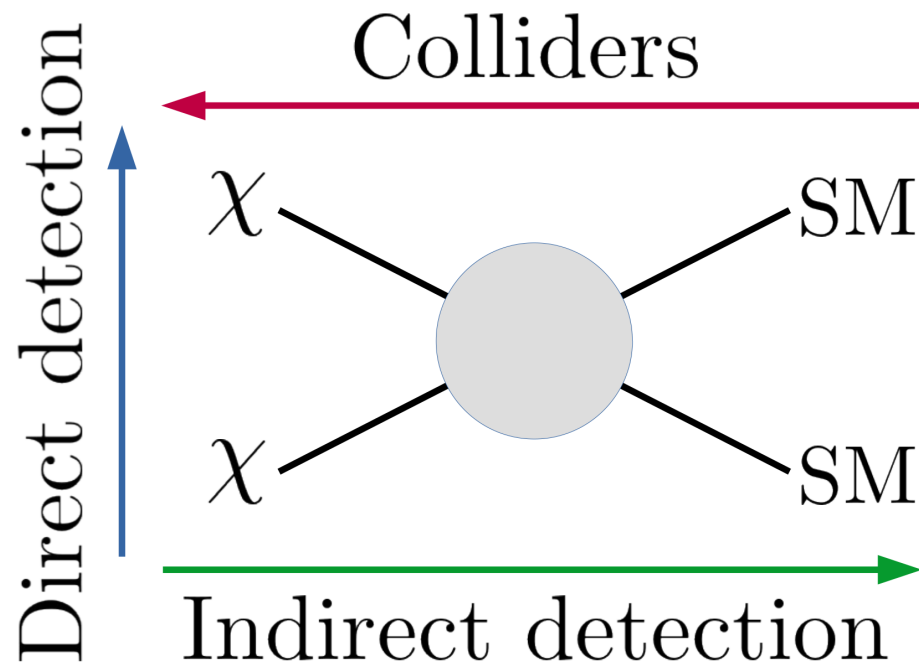
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- Gravity?
- Why Three Generations?
- Why is the Higgs so light?
- Neutrino Masses?
- Why Matter over Anti-Matter?
- Dark Matter? Dark Energy?
- Origin of the Universe?

...

IV. Dark Sector

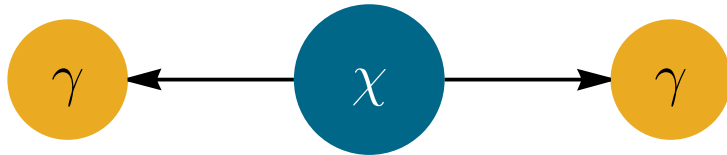
How to Search for Dark Matter



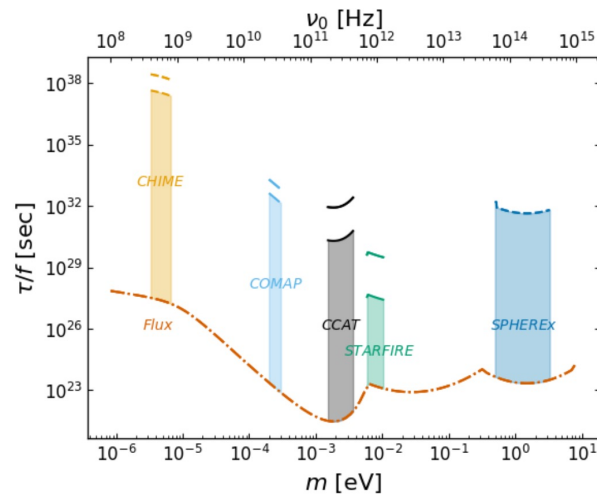
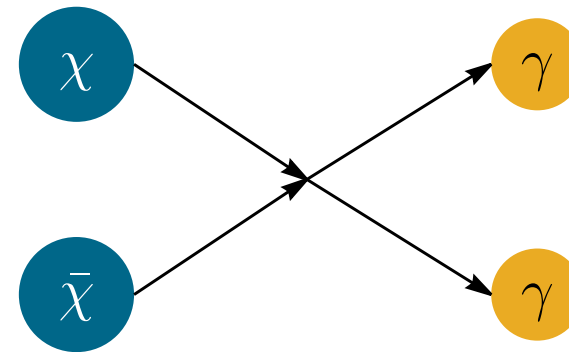
Fundamental Physics Targets

Line-Intensity Mapping Searches for DM couplings to light

DM Decay



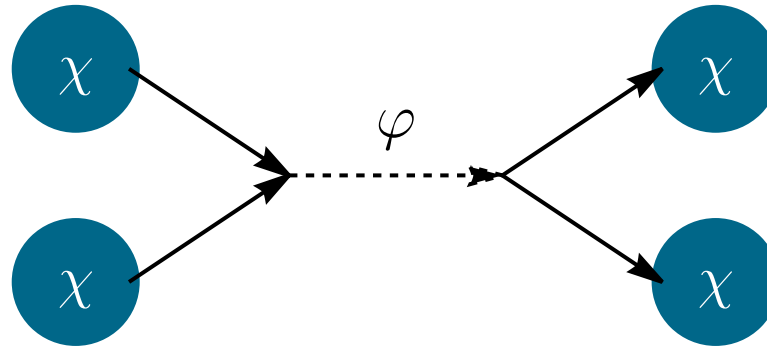
DM Annihilation



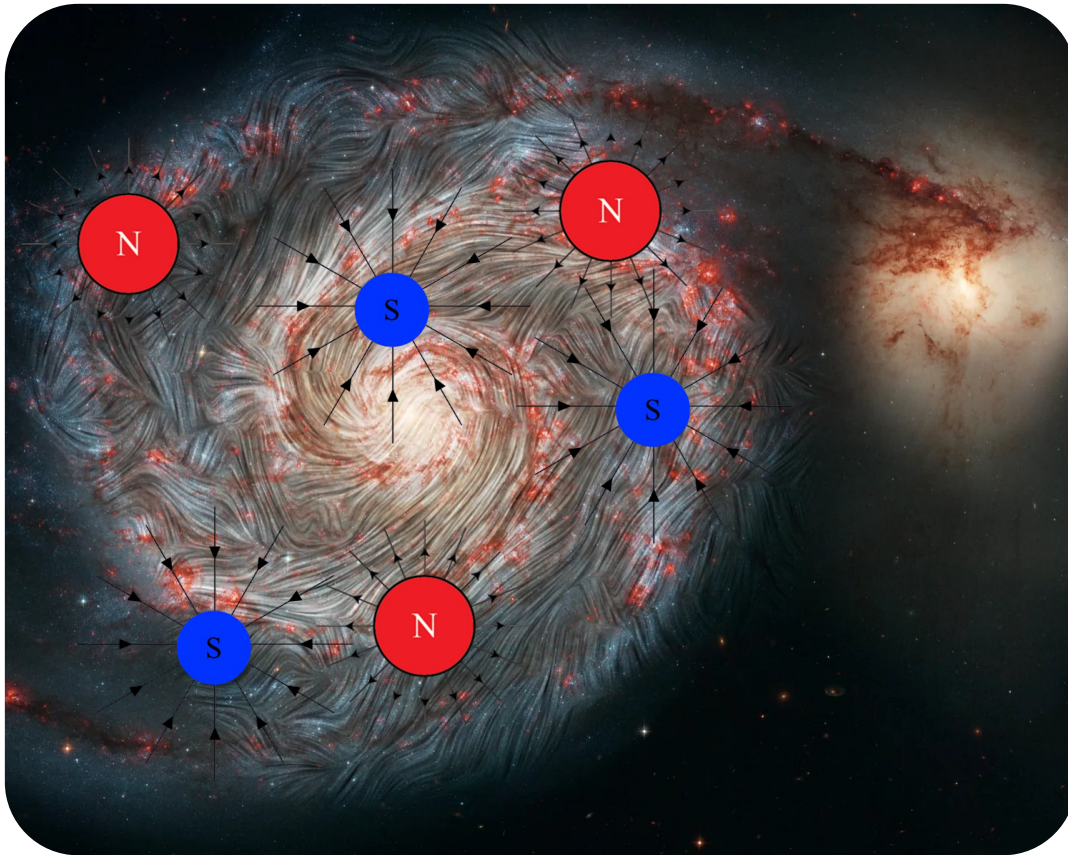
Fundamental Physics Targets

CMB/LSS Searches for long-range dark fifth forces

DM Self – Interaction



Magnetic Monopoles



[Marius Kongsore (grad. Student), Chris Dessert, **CCS**, (in prep)] ³³

Conclusions

- Cosmological Observations can probe a plethora of new physics beyond the Standard Model
 - 1) The Dark Sector (decaying DM, new light species, dark fifth forces)
 - 2) Origin of the Universe (parity violation, Higgs physics, etc)
- There are also lots of astrophysical objects to learn from as well!
 - 1) Active Galactic Nuclei
 - 2) High-Energy Astrophysical Neutrinos
 - 3) Compact Binary Coalescences....

Thank you.

